Routing Protocols — Guided Notes ANSWER KEY

Unit 1.8.2 | CompTIA Network+ N10-009 Obj. 1.8 | N10-008 1.8

For instructor use / distribute after students complete the notes.

Question 1

Distance vector protocols share _______ with directly connected neighbors at regular intervals. Routers only know _______ and _______ to each destination — not the full topology. Link state protocols share _______ with all routers in the area, so every router builds _______.

Answer: Their entire routing table; direction and distance (hop count or metric); Link State Advertisements (LSAs); an identical map of the full network topology.

Question 2

RIP uses _______ as its only metric. Its maximum is _______ hops — a destination _______ hops away is considered unreachable. RIP sends full table updates every _______ seconds, which causes slow _______.

Answer: Hop count; 15; 16; 30; convergence.

Question 3

OSPF uses _______ as its metric, derived from _______. This means OSPF prefers _______ links over _______ links, unlike RIP which only counts _______.

Answer: Cost; interface bandwidth; faster (higher bandwidth); slower; hops.

Question 4

In OSPF, _______ is the backbone area that all other areas must connect to. Using areas limits the scope of _______ flooding and _______ calculations, allowing OSPF to scale to large networks.

Answer: Area 0; LSA; SPF (Shortest Path First).

Question 5

On an Ethernet segment with multiple OSPF routers, a _______ (DR) and _______ (BDR) are elected. All other routers form adjacencies only with these two. This reduces the number of _______ relationships from exponential to manageable.

Answer: Designated Router; Backup Designated Router; adjacency/neighbor.

Question 6

EIGRP's DUAL algorithm maintains a _______ — a pre-calculated backup route. If the primary route fails and one exists, EIGRP installs it _______ with no recalculation. EIGRP only sends updates when _______, making it bandwidth-efficient.

Answer: Feasible successor; immediately (sub-second); something changes (triggered updates).

Question 7

BGP is classified as a _______ protocol. It routes between _______ (ASes), each identified by an _______. _______ BGP runs between different ASes (e.g., your router to your ISP). _______ BGP runs within the same AS.

Answer: Path vector; Autonomous Systems; AS number; eBGP (external); iBGP (internal).

Question 8 — Real World Application

REAL WORLD: You inherit a network with three sites running RIP. Users at Site C (which is 14 hops from Site A) report they can reach Site A fine, but a new Site D has been added 2 hops beyond Site C. Users at Site D cannot reach Site A at all. Explain why and what you would do.

Answer: Site D is 16 hops from Site A (14 + 2), which exceeds RIP's maximum of 15 hops. RIP treats the destination as unreachable and does not install the route. Fix: replace RIP with OSPF or EIGRP, which have no practical hop count limit. OSPF uses cost (bandwidth-based) and scales to any size network. This is also an opportunity to review whether the network design can be optimized to reduce hop count.